

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	SECTION:	CPE21S1
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SEATWORK 8.1 Using Sorting Algorithms 2

1. PROCEDURE OUTPUT

Answer the following questions:

1. Explain the quick sort, shell sort and merge sort types of sorting algorithms.

- Quick Sort:**
 You pick one number from the list and call it the "**pivot**." Then you move all the smaller numbers to one side of it, and all the bigger numbers to the other side. Now the pivot is in its correct final spot. You do the same thing again on the left group and the right group, smaller and smaller, until everything is sorted.
- Shell Sort:**
 Is an improved version of insertion sort that compares numbers separated by a gap distance instead of adjacent elements. As the gap slowly shrinks to 1, distant numbers quickly move to their correct positions, making final sorting much faster.
- Merge Sort:**
 Merge Sort divides an array into smaller parts until each part only contains one element then it combines the parts until it becomes sorted or organized. Basically, Merge Sort splits the value into smaller parts then compares and combines them and just repeats the process until it becomes a completely sorted array.

2. Give simple sample programs in C++ that uses the above sorting algorithms. Use a user input of 10 integer values in your example elements in an array to be sorted. Explain how the programs work.

1. Quick Sort - Temperature Monitoring System

Problem:

Create a C++ Temperature Monitoring System for a computer or machine. The system receives 10 integer temperature readings from sensors and stores them in an array. Use the Quick Sort algorithm to arrange the 10 temperature readings from lowest to highest. The sorted values will help the user identify the lowest and highest temperature recorded.

Example input:

35 28 40 31 25 38 30 42 27 33

Sorted result:

25 27 28 30 31 33 35 38 40 42

How the program works:

The program asks the user to enter 10 temperature readings. These values are stored in an array, and Quick Sort repeatedly selects a pivot and separates smaller and larger values until the array is sorted. Finally, the system displays the temperature readings from lowest to highest.

2. Shell Sort - Smart Parking System

Problem:

Create a C++ Smart Parking System that receives the distance of 10 vehicles from a parking sensor. The system accepts 10 integer distance values and stores them in an array. Use the Shell Sort algorithm to arrange the distances from the nearest vehicle to the farthest vehicle.

Example input:

25 10 40 15 30 5 50 20 35 45

Sorted result:

5 10 15 20 25 30 35 40 45 50

How the program works:

The program receives 10 distance readings from the parking sensors. Shell Sort compares values that are separated by a certain gap and gradually reduces the gap until the values are completely sorted. The system then displays the vehicles according to their distance from the parking area or entrance.

3. Merge Sort - CPU Temperature Monitoring App

Problem:

Create a C++ CPU Temperature Monitoring System that receives temperature readings from 10 computer sensors. The program stores the 10 integer readings in an array and uses the Merge Sort algorithm to arrange them from lowest to highest.

Example input:

55 42 70 48 65 38 80 50 60 45

Sorted result:

38 42 45 48 50 55 60 65 70 80

How the program works:

The program first receives 10 CPU temperature readings and stores them in an array. Merge Sort divides the array into smaller parts, sorts those parts, and then merges them back together in the correct order. At the end, the system displays the temperature readings from the lowest to the highest.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.